EARLY

Early Assisted Reading and Literacy for Youth — Project Brief for Claude

■ This document is the single source of truth for the EARLY project. Read it fully at the start of every session before responding to any request.

Who

Gregory — 40-year-old reading teacher finishing a CS degree at UCLA (pure CS, genomics technical breadth). Trained at Lindamood-Bell. Solo founder. Building in Cursor with Claude as architecture and problem-solving support. Will use the app with his own students to collect the first training data.

What

EARLY is an adaptive early reading platform with three eventual products:

Product	Audience	Status
EARLY Student	K-5 learners	Active development — phoneme practice app working
EARLY Teach	Classroom teachers	Not started — teacher phoneme targeting dashboard
EARLY Assess	Reading specialists	Not started — automated diagnostic + IEP reports

All three products share one engine: a DSP pipeline (FFT, mel filterbank, MFCC extraction) and an online-learning neural network (13 inputs, 16 ReLU, 8 ReLU, 6-class softmax). The student app trains the model. The teaching tool steers focus. The assessment platform reads the accumulated signal.

Curriculum Scope & Sequence

Informed by Lindamood-Bell / Orton-Gillingham. See, hear, manipulate modalities only — kinesthetics excluded based on classroom evidence. Proprietary scope and sequence.

Unit	Name	Word Type	Text?	Notes
1	Alphabet & Shorts	CVC only	No	Stages 1-3 mastered together. All 26 phonemes assessed every sessio
2	Blends, Digraphs & Vowel Teams	Single syllable	No	Blends first (no new concept), then consonant digraphs (introduces tear
3	Irregular Sight Words	Single syllable	Yes — begins	Irregular words only — no decodable words. Most frequent first. Short b
4	Multisyllabic Words	Multi-syllable	Yes	Open/closed syllable rule named explicitly. Student arrives with closed s

Mastery threshold: 90% accuracy across 2 consecutive sessions. 2-3 words per target phoneme per session. Unit 3 uses whole-word accuracy (speech-to-text), not phoneme analysis.

Tech Stack

Layer	Technology	Detail
Frontend	Vanilla HTML/JS	Single file. No framework. ES5-compatible for iframe safety.
DSP pipeline	Pure JS (no deps)	FFT, 26-filter mel filterbank, DCT-II, 13 MFCCs, ZCR, RMS, centroid.
Neural net	Pure JS (no deps)	13->16->8->6 softmax. Glorot init. Backprop + batch SGD. lr=0.02.
Speech-to-text	Web Speech API	webkitSpeechRecognition. Chrome/Safari only. Word-level matching.
Backend	Python + FastAPI	POST /api/samples, GET /api/model, POST /api/sessions.
Database	PostgreSQL	mfcc_samples, sessions, model_versions tables.
Retraining	scikit-learn MLPClassifier	Nightly cron at 2am. Min 20 samples before retrain runs.
Server	IONOS VPS	Ubuntu 22.04 LTS. Nginx reverse proxy. Systemd for FastAPI.
SSL	Let's Encrypt / certbot	Auto-renewing. Required for iPad microphone access.
DNS	GoDaddy	ns75/ns76.domaincontrol.com. A record: early.gregtutors.com.
Domain	early.gregtutors.com	Frontend + backend on same domain. No CORS needed.
Deploy	Git pull + systemctl restart	SSH in, run /app/deploy.sh. No CI/CD yet.
Device	iPad (Safari)	Primary student device. iOS Safari only — WebKit engine.

Server Architecture

Everything on one IONOS VPS. Vercel is cancelled.

Path	Serves
/	Static frontend — /app/early/dist
/api/	FastAPI on port 8000 (proxied by Nginx)
/model/	weights_latest.json — static file, loaded at session start

Data Flow

- Student speaks a word. MediaRecorder captures audio (audio/mp4 on iOS, audio/webm on desktop).
- Client DSP pipeline: FFT -> mel filterbank -> log -> DCT-II -> 13 MFCCs per frame.
- Voiced frames isolated by RMS threshold. Middle 60% = nucleus. 13 MFCCs averaged = feature vector.
- Client neural net runs inference on feature vector. Probability bars shown (vowel groups only).
- Web Speech API transcribes. Match = exact, substring, or Levenshtein <= 1.
- Correct match + vowel word: sample added to sessionBuffer (not sent yet).
- Session end: one POST /api/samples sends entire buffer. One write per session, not per word.

• Nightly 2am: retrain.py loads all correct samples from PostgreSQL, retrains MLPClassifier, saves weights JSON.

• Next session start: client fetches /api/model and loads updated weights into JS neural net.

Critical iOS / iPad Constraints

■ All browsers on iPad are Safari under the hood (Apple WebKit requirement). Test only in Safari. Chrome and Firefox on iOS behave identically.

• MediaRecorder: use audio/mp4 on iOS, audio/webm on desktop. Detect with MediaRecorder.isTypeSupported().

• SpeechRecognition: must be triggered by a direct user gesture (tap). Do not call .start() programmatically.

• Microphone access requires HTTPS — self-signed certs do not work, must use Let's Encrypt.

• Touch targets minimum 64px for young students. Tap-to-speak button should be large and central.

• Portrait orientation. Target word large at top, button large at bottom.

• PWA: add manifest.json and apple-mobile-web-app-capable meta tag for home screen install.

Current State & Immediate Priorities

Area	Status	Next action
Phoneme app	Working	iOS Safari testing, touch target sizing, pedagogy pass on feedback text
VPS	Reinstalling	Ubuntu 22.04, Nginx, PostgreSQL, FastAPI, certbot — see setup guide
DNS	Pending	Add A record for early.gregtutors.com in GoDaddy
Retraining pipeline	In progress	Migrate from Vercel blob to PostgreSQL on VPS, fix to nightly cron
Session logging	In progress	Batch writes per session working — wire to VPS endpoint
Sight words	Not started	Unit 3 — after VPS and Unit 1-2 are solid in classroom
Short books	Not started	Unit 3 — after sight words
EARLY Teach	Not started	After classroom validation of student app
EARLY Assess	Not started	After sufficient session data accumulated

Working Notes for Claude

• Gregory is a reading specialist and CS student — match technical depth accordingly. He knows both domains well.

• He prefers direct answers over hedged ones. If something is the right call, say so.

• Deliver final documents as PDFs, not .docx.

• When generating PDFs, use ReportLab. Avoid HTML tags in code blocks inside Paragraph() — ReportLab parses them.

- He works in Cursor on Windows with an IONOS VPS and deploys via SSH.
- Primary test device is an iPad. Always consider iOS Safari constraints when suggesting frontend solutions.
- The neural net and DSP pipeline are intentionally implemented in pure JS with no dependencies — keep it that way for the client.
- EARLY is a solo project targeting classroom use within one month. Keep scope tight and solutions practical.
